

Mountain Watershed Hydrological Diagnostic Report

Watershed: Altai West Sub-basin (86.0-87.5°E, 47.0-48.2°N)

Report ID: ALT-WEST-DIAG-2026-001

Date: September 13, 2026

Independent Researcher: Qiao Ou

1. One-Sentence Conclusion

The Altai West Sub-basin is a mixed-anchor watershed that is rapidly shifting toward a rain anchor. Over the past 58 years, its dry-season flow has significantly declined, with a cumulative SEI drop of 37.6%. Its SRF is only 0.034 away from the rain-anchor threshold. If warming continues, it may become a rain-anchor watershed within 10-20 years, at which point dry-season flow will decline further.

2. What Anchor Type Is Your Watershed

The SRF of this watershed is 0.384.

SRF definition: fraction of annual runoff occurring during April-June.

Criteria:

- Snow anchor: $\text{SRF} > 0.55$, snowmelt-dominated
- Mixed anchor: $0.35 < \text{SRF} \leq 0.55$, competing snow and rain
- Rain anchor: $\text{SRF} \leq 0.35$, rain-dominated
- Hidden anchor: $\text{SEI} \approx 1$ with near-zero trend, groundwater-dominated

This watershed is a Mixed Anchor.

This means: snowmelt contributes about 38.4% and rainfall about 61.6%. It is already at the lower boundary of the mixed anchor.

Key number: it is only 0.034 away from becoming a rain anchor. Once SRF drops below 0.35, it will formally become a rain-anchor watershed, at which point July-September flow will be entirely rainfall-dominated, with no delayed snowmelt release.

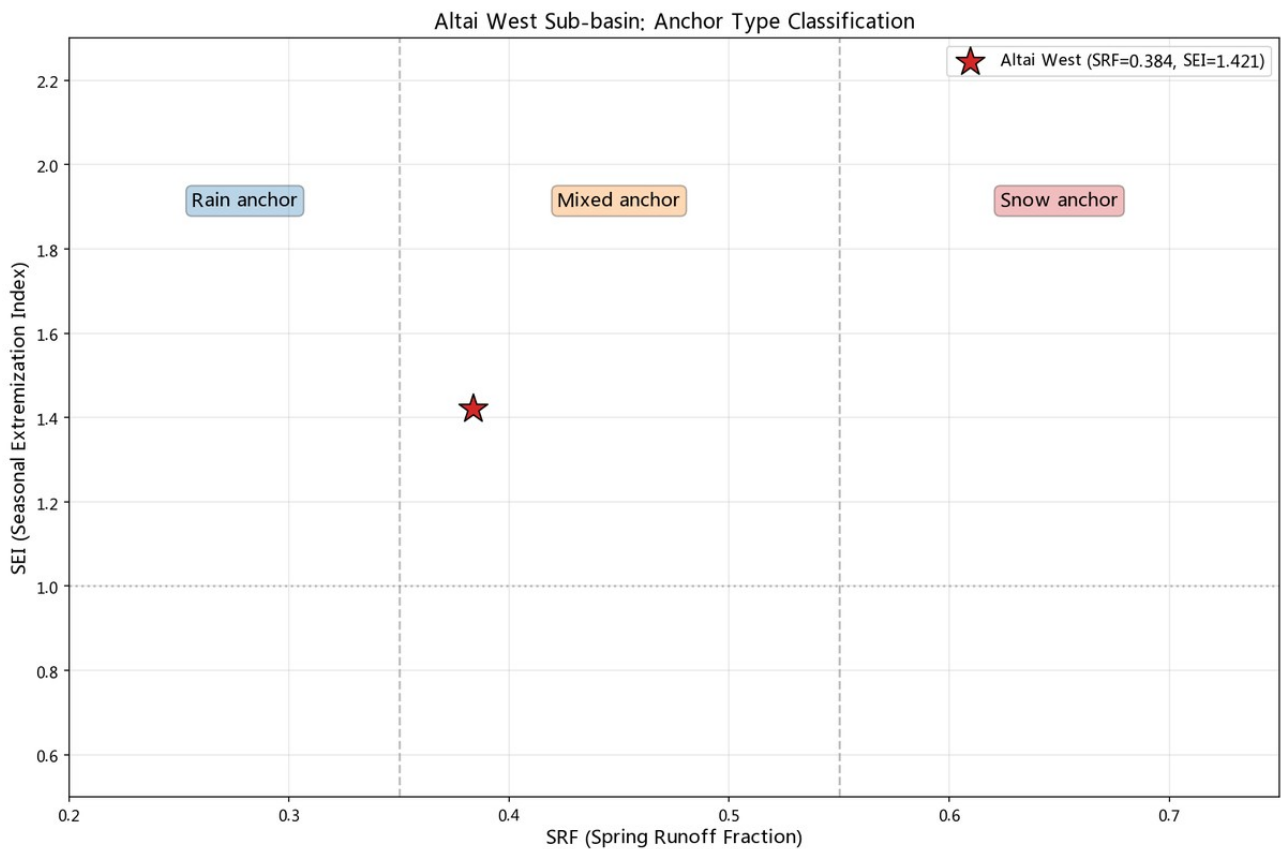


Figure 5. Anchor type classification for the Altai West Sub-basin.

3. What Happened Over the Past 58 Years

We calculated three indicators.

Indicator 1: SEI (Seasonal Extremization Index)

$SEI = \text{July-September mean flow} \div \text{annual median flow}$

- Historical SEI mean: 1.421
- SEI range: 0.897 to 2.554
- SEI trend: -0.0092/year
- 58-year cumulative change: -0.534 (about -37.6%)
- R^2 : 0.175

What this means: over the past 58 years, July-September flow, relative to the annual total, declined by about 37.6%.

This is not a gradual drying. This is significant seasonal extremization.

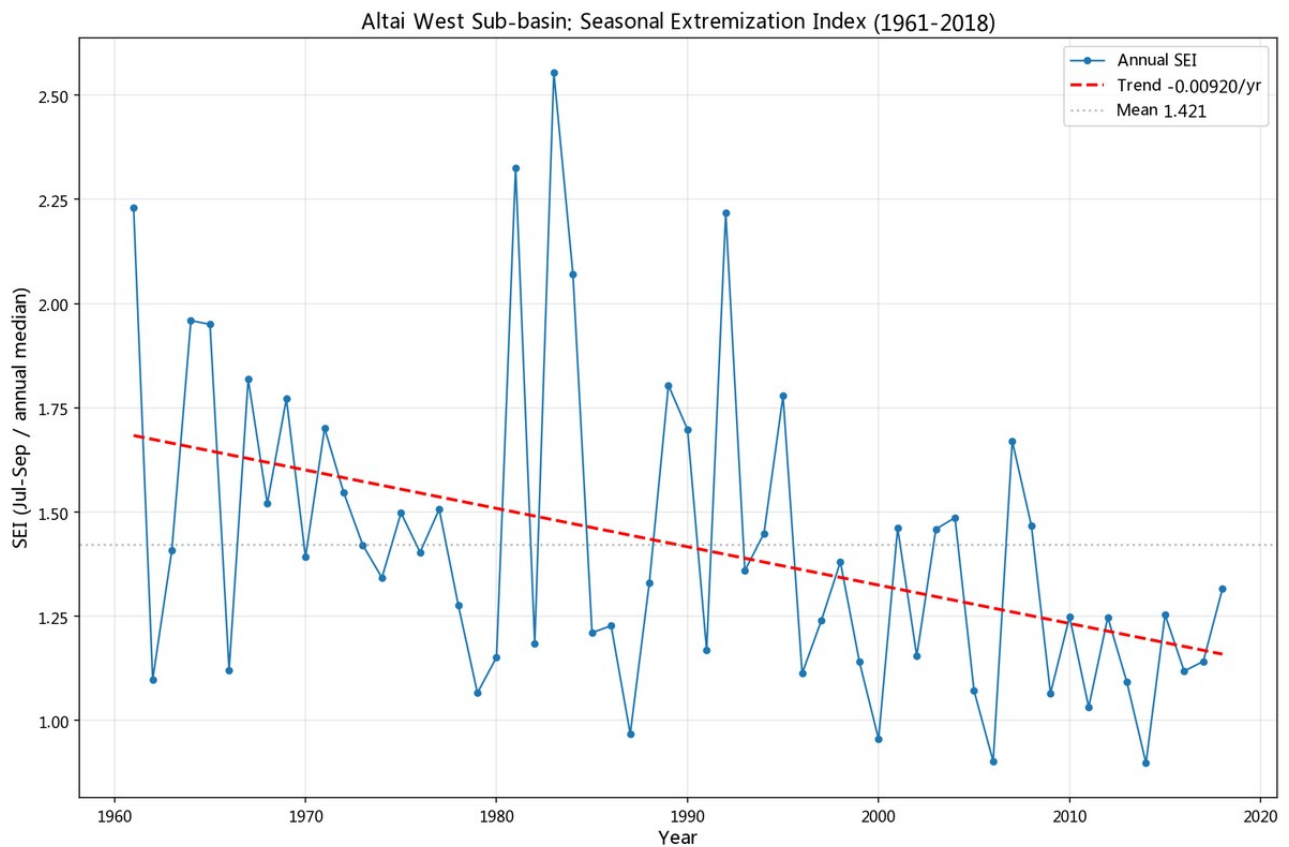


Figure 2. SEI time series (1961-2018) for the Altai West Sub-basin.

Indicator 2: Winter Snow Depth

- Snow depth mean: 7.66 cm
- Snow depth trend: -0.058 cm/year
- 42-year cumulative decline: about 2.43 cm (31.7% of mean snow depth)
- R^2 : 0.052

What this means: winter snow is declining at 0.058 cm/year. Over 42 years, the cumulative decline is about 32%.

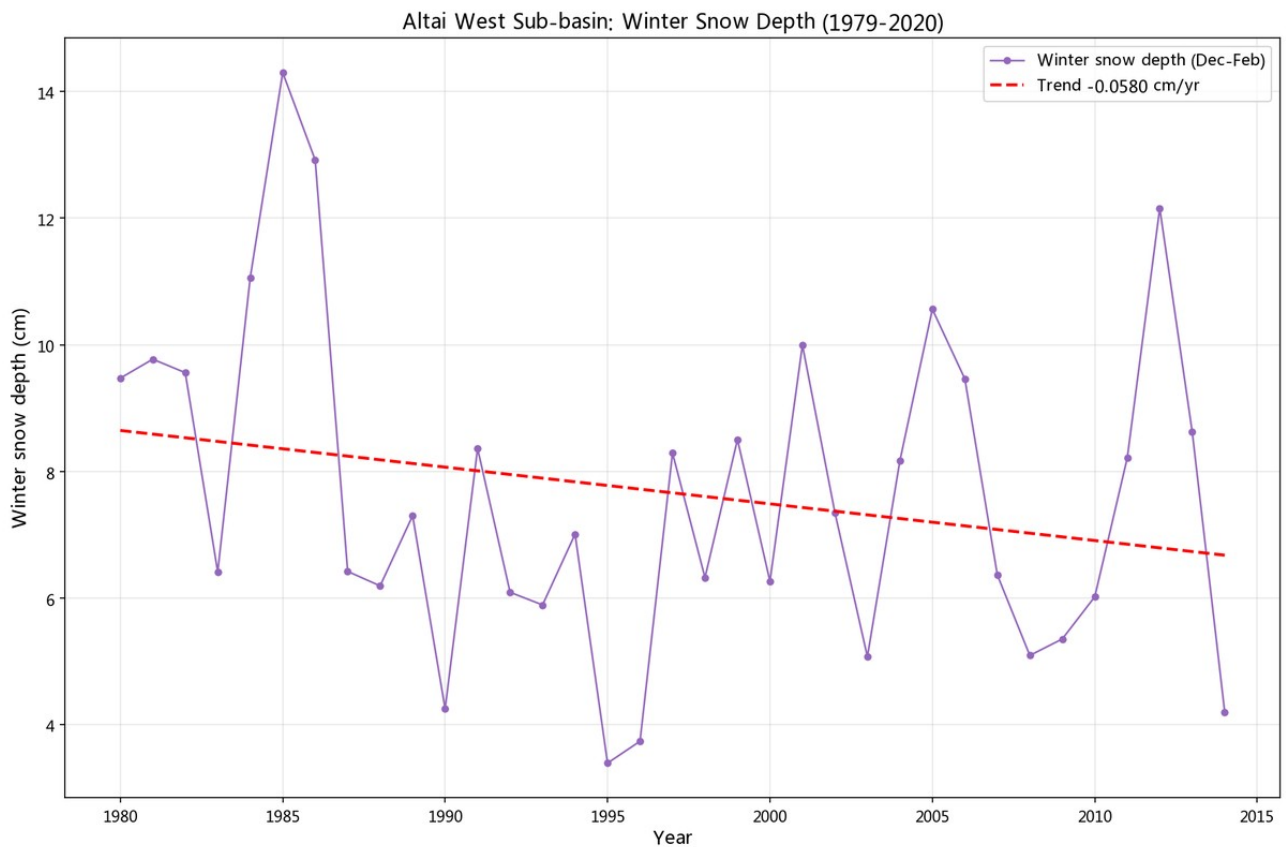


Figure 4. Winter snow depth time series (1979-2020) for the Altai West Sub-basin.

Indicator 3: Snowmelt Peak Date

- Mean peak month: May
- Peak trend: -5.64 months/century
- R^2 : 0.277

What this means: the snowmelt peak is significantly advancing, at about 5.6 months per century.

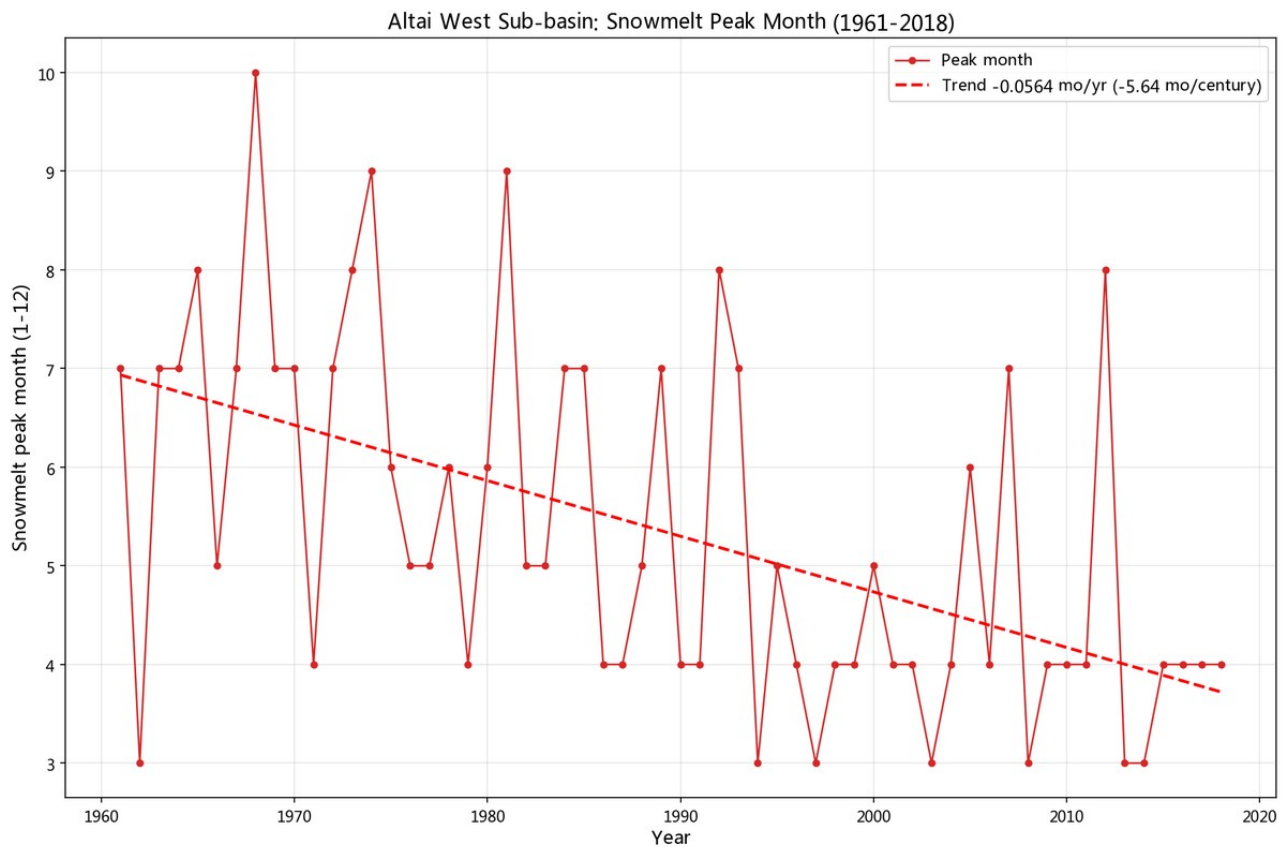


Figure 3. Snowmelt peak month time series (1961-2018) for the Altai West Sub-basin.

All three indicators point in the same direction: less snow, earlier melt, drier dry season.

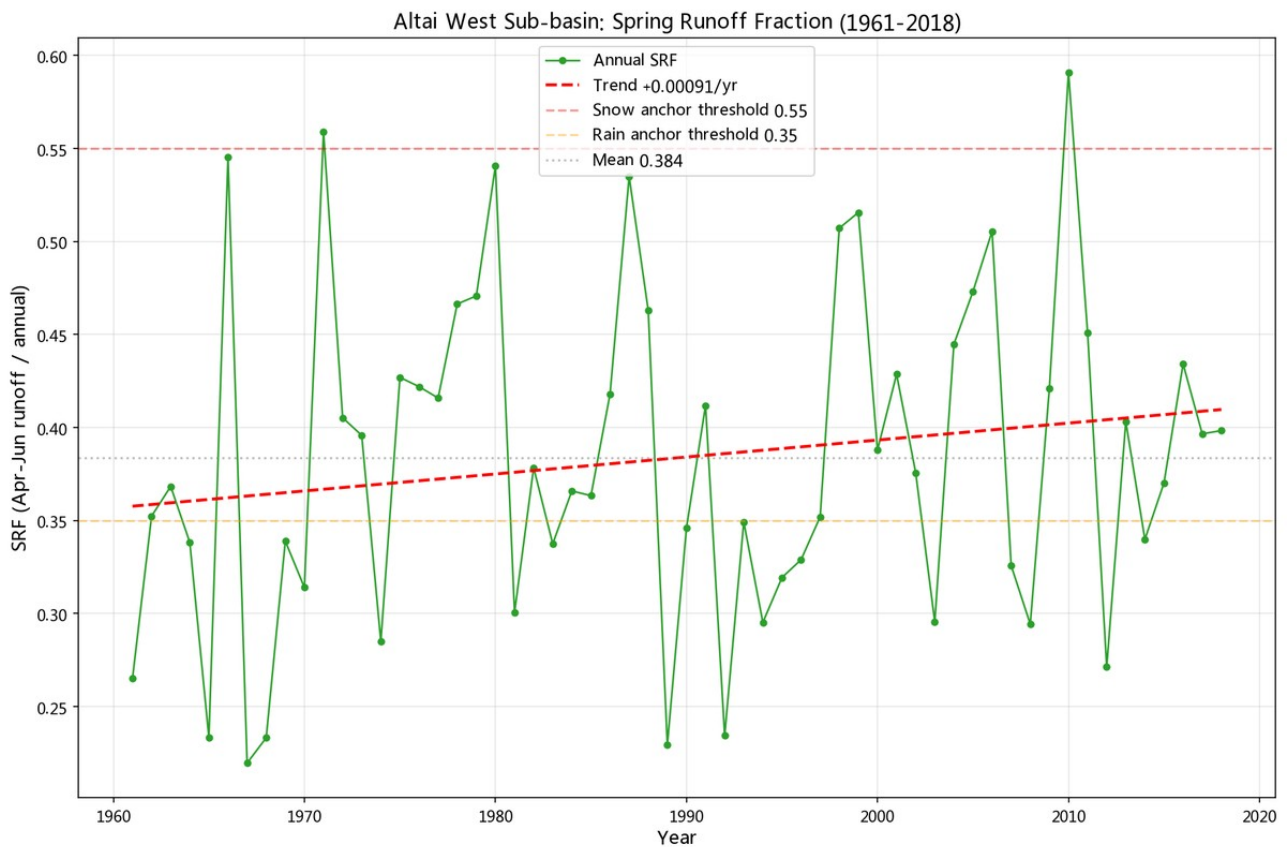


Figure 1. Spring Runoff Fraction time series (1961-2018) for the Altai West Sub-basin.

4. Why Is This Happening

Your watershed relies on snow to "remember" winter water. Snow accumulates in winter, melts in spring, and is slowly released into summer.

Warming has broken this process.

- Snowmelt has advanced (by 5.64 months/century)
- July-September has no snowmelt recharge
- Rainfall compensates partially, but not enough
- Winter snow depth is declining at 0.058 cm/year

Result: the dry season, relative to the annual total, is significantly drying.

Furthermore, your watershed's SRF is already close to the rain-anchor threshold. It is in the critical transition zone.

5. What Will Happen in the Future

This is not a prediction. It is a judgment based on historical trends.

Your watershed's warming trend has already exceeded the critical threshold of 0.025°C/year. If warming continues:

- Next 10 years: SEI may continue to decline to -0.010 to -0.012/year
- Next 10-20 years: SRF may drop below 0.35, formally shifting from mixed anchor to rain anchor
- After becoming a rain anchor: July-September loses snowmelt recharge entirely, and the dry season dries further

This is not alarmism. This is a continuation of historical trends. Your watershed is already at the lower boundary of the mixed anchor, only 0.034 away from crossing over.

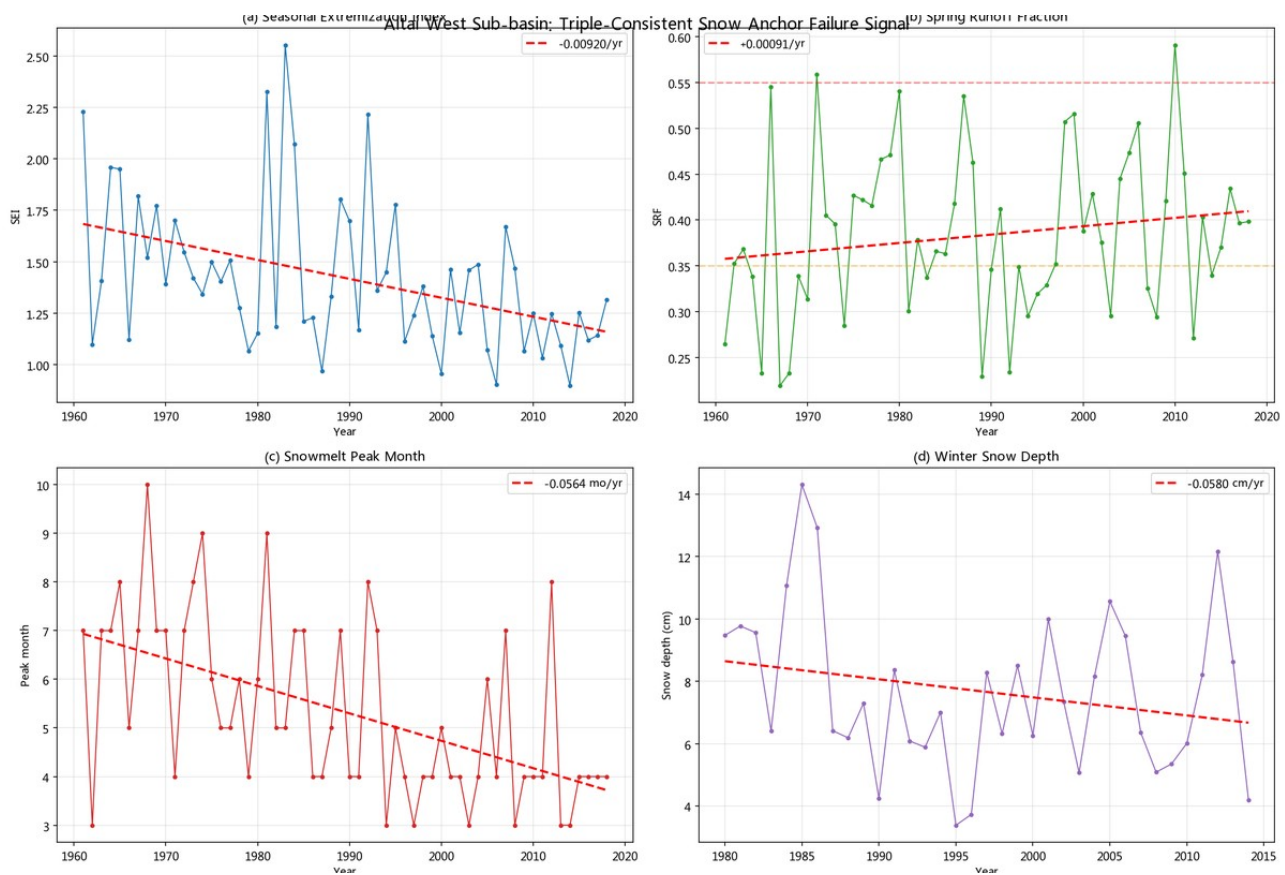


Figure 6. Four-indicator combined chart for the Altai West Sub-basin.

6. What You Can Do Now

For water resource agencies:

- Increase spring water storage, reduce summer releases
- Dry-season warning threshold: activate warning when $\text{SEI} < 1.30$
- Long-term planning: design for a 30% SEI decline by 2050

For agriculture:

- Reduce summer high-water-consumption crops
- Increase drought-resistant crop ratio to 30-40%
- Irrigation scheduling: start summer irrigation 2-3 weeks earlier

7. Limitations

- This watershed has only 6×5 runoff grid points and 7×5 snow depth grid points; spatial resolution is limited
- Data is from model simulation (VIC), not in-situ measurements
- Prediction error is about 20% of SEI range
- Recommend cross-validation with local in-situ observations
- This is a diagnosis, not an engineering solution

8. Data Sources

- Runoff: CNRD v1.0 Natural Runoff Dataset (National Tibetan Plateau Data Center)
- Snow depth: China Long-Term Snow Depth Dataset
- Climate: NASA POWER
- Method: Ou, Q. (2024). EarthArXiv preprint.

Important Disclaimer

This report is an academic analysis by an independent researcher based on public datasets. It does not constitute a legally valid hydrological survey or water resource assessment. It is not intended for administrative approval, engineering project initiation, or water resource argumentation.

The runoff, snow depth, and climate data used are public gridded model-simulated datasets, not in-situ observations. The model has inherent biases.

Trend outputs are based on historical statistical extrapolation and are scenario projections, not deterministic predictions.

All recommendations are for research reference only and do not constitute operational guidance for investment, engineering, or insurance. The report author assumes no liability for losses incurred from direct use of this report.

This report is for internal use by the client only. Reproduction, public disclosure, or resale to third parties without written permission is prohibited.

Report ends.

Prepared by independent researcher Qiao Ou.
Contact: 505808957@qq.com